Organized Thinking Through the Study of Fundamental Systems

Senior Graduation Narrative

Johnston Center for Integrative Studies • University of Redlands

I graduated from high school in June 2021, at the tail end of a pandemic that had kept all of us inside, isolated, and at a distance from the kinds of formative experiences that are supposed to define those final years before adulthood. I arrived at the University of Redlands having spent much of my last year in high school in front of a screen, and I was not entirely sure what college was supposed to feel like. Before I ever stepped foot in a classroom, the university's Outdoor Programs orientation gave me my first answer. Incoming students were placed into small groups, led by upperclassmen, and sent somewhere unfamiliar. My group went to Catalina Island. We backpacked. We swam. We hiked. We spent several days getting to know each other with nothing in particular to do except be present. It sounds simple, but for someone coming out of two years of Covid-era isolation, it was genuinely transformative. I met some of my closest friends on that trip. More than that, it taught me that community is something you build through shared experience, a lesson I would carry into everything I did at Redlands.

The emphasis I designed for myself at the Johnston Center for Integrative Studies is titled *Organized Thought Through Fundamental Systems*. Johnston is a program within the University of Redlands that allows students to design their own course of study in place of a traditional major, working closely with faculty advisors to build a coherent, self-directed curriculum. My emphasis is an interdisciplinary program built around physics, computer science, and philosophy; three fields that, taken together, offer a rigorous and interconnected way of understanding how the world works at its most foundational level. The animating question behind all of it is this: what are the underlying principles that govern complex systems, and how can understanding them allow us to make meaningful changes in the world?

Physics gave me the first layer of that answer. Working through three semesters of General Physics and then Electronic Applications, I developed not just knowledge of the laws governing the physical world but a habit of thinking; methodical, honest, willing to follow an argument wherever it leads, even when the destination is inconvenient. Physics does not reward wishful thinking. It taught me to construct rigorous arguments from first principles and to accept when evidence contradicts expectation. That discipline became the foundation for everything else.

Computer science gave me a complementary kind of rigor: the logic of building things. Aside from a two week long summer camp I attended in middle school at my local junior college, where I learned how to draw with a little turtle using code, I had never programmed before I enrolled in Introduction to Programming in my first semester. I remember the specific feeling of writing my first program. It was the first time I felt as though I was speaking the language of my own brain fluently. That moment reoriented my sense of what I wanted my education to be. I moved through data structures and

object-oriented programming, web application development, mobile programming, and eventually game programming and artificial intelligence, each course deepening my understanding of how software systems are designed. But it was at Columbia University's Fu Foundation School of Engineering and Applied Science, where I have spent two years completing a second undergraduate degree in Computer Science through the 3-2 Engineering Combined Plan, that this understanding matured into something more rigorous. Columbia's engineering curriculum taught me the formal mathematics of how data is structured, manipulated, and transmitted. I came to understand that computer science is not just programming, but the intricate and technical study of computation, design, and the theory of information.

Philosophy provided the perspective that unifies my studies in physics and computer science. At Redlands, formal symbolic logic gave me precise tools for evaluating arguments, while my study of Confucianism introduced an ethical tradition grounded in relationships and duty rather than abstract universal principles, challenging me to ask not only can this be built, but should it be, and for whose benefit? Those two threads ran alongside each other for years without fully converging. It was not until I enrolled in Cognitive Science and Philosophy at Columbia that I found the bridge between them. That course sits at the intersection of philosophy, psychology, and computer science, examining foundational questions about the nature of mind: What is cognition? Is the mind like a computer? Does thinking depend on internal representations, or on embodied engagement with the world? Working through classical computationalism, theories of consciousness, and the philosophy of artificial intelligence, I finally had a framework that let me bring the logical precision of my computer science training into direct conversation with the philosophical questions I had been carrying my whole life. The course did not resolve those questions, good philosophy rarely does, but it gave me a language for holding them together, and it clarified what I believe my work in technology is ultimately about; building systems that account for the full complexity of the minds that use them.

Beyond my core emphasis, I pursued a deliberately wide curriculum. Spanish language study reconnected me with my cultural heritage as my family roots run deep in Spanish culture, and returning to the language felt like returning to something I had let drift. Piano, which I returned to after years away, reminded me that discipline applied to something beautiful is its own reward, entirely separate from utility or career. My political science and public policy courses, spanning American politics, California governance, policy analysis, and an urban planning course I took at Columbia, gave me a genuine appreciation for how civic institutions are built and sustained, and for the people who dedicate themselves to that work.

Two courses stand out above the rest. Equal Protection, my first constitutional law course, was one of the most enjoyable academic experiences of my life. Reading and arguing Supreme Court opinions and sitting with the ambiguity, the competing interpretations, the weight of what those decisions meant for real people taught me a kind of close reading I had never practiced before. That love of legal argument carried me directly to Salzburg, Austria, where I spent a month abroad in a course on Comparative Judicial Systems: Human Rights. We lived in a beautiful historic city

studying how European courts approach the same fundamental questions of human dignity that American courts grapple with. It was, in every sense, an education: in law, in culture, and in how differently people across the world understand what it means to be treated fairly.

I utilized my time at Redlands outside of the classroom to learn who I was as a person. During my first weeks of college, I ran for, and was elected, the freshman senate representative for the Associated Students of the University of Redlands, our student government at the time. And throughout my time at Redlands, I never left the leadership role. At the end of my first year I was awarded "Outstanding Achievement by a First Year", described as the "recognition of a first year student who has exhibited exemplary qualities of leadership and earned a distinguished record of campus involvement". I was also selected as a member of the Maroon and Grey Student Ambassadors, a group that represents the university to its broader community and alumni network, where I participated and volunteered for events for the rest of my time at Redlands. I also completed a formal tutor certification course and spent time outside of class tutoring fellow Redlands students in computer science. It was some of the most fulfilling work I did at the university. Teaching something you love to someone who needs it has a way of deepening your own understanding in ways that studying alone never quite does. I continued to be a part of the student government through my sophomore year. In my last year at the university, when the student government was in a hiatus, I founded a new student organization, Doing Activities With Great Spirit, dedicated to campus programming and event planning that grew into a formally recognized body alongside the student government that was allotted a budget drawn from student activity fees. We organized events, built student sections at athletic games, and collaborated with other campus organizations. The university recognized us for our contribution to student engagement.

Living in New York City has been its own education. Moving across the country from my home and everything familiar in California and landing in one of the most demanding intellectual and cultural environments in the world changed something in me that is difficult to name precisely but impossible to ignore. I have grown here in ways that have less to do with coursework than with the accumulated experience of navigating a new city, building a new community, and discovering that I am capable of more than I knew. I traveled abroad the summer before arriving and then stepped almost immediately into the pace and scale of New York. What I have come away with is a changed relationship to the world itself. I no longer think of home as a fixed place. I think of the world as somewhere I can move through, contribute to, and belong in. That shift in perspective may be the most durable thing my education has given me.

What I carry out of this education, across both institutions, across five years, is harder to summarize than a transcript. I have technical depth: I can program across multiple languages and paradigms, build and deploy web and mobile applications, analyze data, and read legal opinions. But more than any individual skill, I have a way of approaching problems. I break systems down to their components, identify the principles governing each, and ask how they might be reorganized to work better. That habit of mind was the premise of my emphasis at Redlands, and two years at Columbia have sharpened it into

something I trust. I am drawn toward work at the intersection of technology and public systems, government technology, civic infrastructure, and the software that shapes how people access services and participate in the life of their communities. I do not yet know exactly what form that work will take. I know that I am prepared to find out.